



The SVG Infinite Plane & The viewBox Lens

In SVG, vector data exists on a boundless Cartesian plane (where $+Y$ goes down). The `viewBox` attribute defines a specific fractional mathematical region to be extracted and projected into the HTML DOM. Anything outside this bounding box is conceptually clipped or ignored during responsive mapping.

∞ Endless Coordinate Space

Coordinates like `x="150"` or `x="-50"` are perfectly valid. The path data mathematically exists here regardless of whether it is visible on the screen.

SVG $(-20, -65)$



width: "100"

Path P_1 (Out of bounds)

🔍 The viewBox Attribute

```
<svg viewBox="0 0 100 100">
```

This dictates: "Start at origin $(0,0)$, and frame a window 100 units wide and 100 units tall." This establishes the *local fractional aspect ratio* (1:1 in this case).

$(0,0)$ Origin

SVG $x = 50$

SVG $x = 100$

X-Axis (∞)

min-x: 0
min-y: 0

height: "100"

SVG $y = 100$

Polygon G_1

Circle C_1

✂ Visual Clipping

Elements intersecting the `viewBox` perimeter (like this circle at $x = 100, y = 100$) are clipped by the DOM's rendering engine. Only the segment inside the $0 - 100$ domain is passed to the CSS scaling engine.

Y-Axis (∞ , Positive Down)